

**Microprocessor Interfacing
and Programming (EE3002)**

Date: September 23rd 2024

Course Instructor(s)

Mr. Maaz Rizvi

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 15

Total Questions: 3

Roll No

Section

Student

Do not write below this line

Attempt all questions

Marks: 5

Question No. 1 (CLO No. 1)

Give the code using 8086 Instruction Set to swap even and odd bits of the number 56H and store it in a new register

Question No. 2 (CLO No. 1)

Give Memory mapping and draw the chip select mechanism for interfacing 256kB of RAM and 128kB of ROM with the 8086 processor. Also, specify the following for both RAM and ROM.

- 1) No. of Address Lines
- 2) No. of Data Lines
- 3) Control Lines

Marks: 5

Question No. 3 (CLO No. 1)

Explain the pipelining effect in the 8086 processor.

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**Microprocessor Interfacing
and Programming (EE3002)**

Date: November 4th 2024

Course Instructor(s)

Mr. Maaz Rizvi

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 15

Total Questions: 3

Roll No

Section

Student Signature

Instructions

Solve the paper on a separate answer sheet

You can use the Instruction Set provided to you on page 2 of the question paper

Attempt all questions

Question No. 1 (CLO No. 3)

Marks: 5

Produce a code in PIC Assembly to add four 8-bit numbers placed at the following locations. Store the higher byte and lower byte in registers named H_BYTE and L_BYTE respectively.

AB C D E F G H

H-BYTE L-BYTE

4C ---- 0x40

3F ---- 0x41

D3 ---- 0x42

A1 ---- 0x43

Question No. 2 (CLO No. 3)

Marks: 5

Assuming that XTAL is 16 MHz, **Produce** a code in PIC Assembly using **Timer 1** to generate a square wave of 40 ms on PORTB, 7. Show the calculations.

$$\frac{1}{4} \times 16 \times 10^6 = 4 \text{ MHz}$$

$$\frac{1}{4 \text{ MHz}} = 0.25 \mu\text{s}$$

Marks: 5 (3+2)

Question No. 3 (CLO No. 2)

a) **Explain** the role of Stack and subroutines in PIC. What happens when the Stack overflows?

b) A developer is using PIC18 microcontroller for a product. The Chip has only 4K on-chip flash ROM. **Explain** which of the two instructions **CALL** or **RCALL** is more useful in programming this Chip.

SHRC
B7

- ADDLW, Literal
- ANDLW, Literal
- SUBLW, Literal
- XORLW, Literal
- MULLW, Literal
- ADDWF fileReg, d
- ADDWFC fileReg, d
- ANDWF fileReg, d
- IORWF fileReg, d
- SUBFWB fileReg, d
- SUBWF fileReg, d
- SUBWFB fileReg, d
- XORWF fileReg, d
- COMF fileReg, d
- DECF fileReg, d
- DECFSZ fileReg, d
- DECFSNZ fileReg, d
- INCF fileReg, d
- INCFSZ fileReg, d
- INCSNZ fileReg, d
- MOVF fileReg, d
- NEGF fileReg, d
- RLCF fileReg, d
- RLNCF fileReg, d
- RRCF fileReg, d
- RRNCF fileReg, d
- SWAPF fileReg, d
- BTG fileReg, d
- IORLW, Literal
- MOVFF fileReg1, fileReg2
- BC
- BNC
- BZ
- BNZ
- BN
- BNC
- BOV
- BNOV
- BRA
- GOTO
- CALL
- RCALL
- RETURN
- NOP
- CLRF
- SETF
- BCF fileReg, bit
- BSF fileReg, bit
- BTFSS fileReg, bit

- BTFSC fileReg, bit
- BTG fileReg, bit
- STATUS
- C - Carry flag
- DC - Digital Carry flag
- Z - Zero flag
- OV - Overflow flag
- N - Negative flag
- CPFSGT fileReg, d
- CPFSEQ fileReg, d
- CPFSLT fileReg, d
- VAR_NAME EQU Literal/Location
- ORG

- TMR0H/TMR1H
- TMR0L/TMR1L
- TMR0IF/TMR1IF
- INTCON
- PIR1

T0CON -

- TMR0ON D7 Timer0 ON and OFF control bit
- T08BIT D6 Timer0 8-bit/16-bit selector bit
- T0CS D5 Timer0 clock source select bit
- T0SE D4 Timer0 source edge select bit
- PSA D3 Timer0 prescaler assignment bit
- T0PS2:T0PS0 D2:D1D0 Timer0 prescaler selector

T1CON -

- RD16 D7 16-bit read/write enable bit
- TICKPS2:TICKPS0 D5:D4 Timer1 prescaler selector
- T1OSCEN D3 Timer1 oscillator enable bit
- T1SYNC D2 Timer1 synchronization
- TMR1CS D1 Timer1 clock source select bit
- TMR1ON D0 Timer1 ON and OFF con

MOVWF W
20
FU
20
21
0A
1

**Microprocessor Interfacing
and Programming (EE3002)**

Date: December 16th 2024

Course Instructor(s)

Mr. Maaz Rizvi

Final Exam

Total Time (Hrs): 3

Total Marks: 50

Total Questions: 5

Roll No

Section

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Instructions

- Attempt all questions
- Answer all questions on the separate answer sheet
- There are 2 pages of the question paper. Please turn over to see the rest of the questions.
- Make sure your code is neatly written. Excessive cutting/overwriting will be penalized.

Question No. 1 (CLO No. 3)

Marks: 3+7

(A) Write three key differences between a microprocessor and a microcontroller. Which one is more suitable for Embedded applications and why?

(B) Draw the memory map to interface 64 KB of RAM and 256 KB of EPROM with 8086 processor. Also include the number of data lines, and address lines for each RAM and EPROM.

Question No. 2 (CLO No. 4)

Marks: 10

Compose a code in PIC Assembly Language for the following task

The CPU continuously performs the task of transferring data from PORTC to PORTD. After the switch SW1 is pressed, the buzzer will start beeping with a frequency of 0.2 kHz. The buzzer will only go off after completing 300 beep cycles or when SW2 is pressed. (You must show your calculations for the timers)

Question No. 3 (CLO No. 4)

Marks: 3+5

(A) Formulate the pros and cons of using C and Assembly Language to program microcontrollers.

(B) Compose a program in C18 that checks the combination of RB0, RB1, and RB2 and performs the operation on A and B according to the following table

	RB2	RB1	RB0	OPERATION
0	0	0	0	ADD A AND B
1	0	0	1	SUBTRACT A AND B
2	0	1	0	MULTIPLY A AND B
3	0	1	1	DIVIDE A by B
4	1	0	0	INVERT A
5	1	0	1	XOR A AND B
6	1	1	0	AND A AND B
7	1	1	1	OR A AND B

0000 0111 $\Rightarrow$ 7

Question No. 4 (CLO No. 4)

Construct a code in PIC Assembly Language for the following problem statement: The CPU constantly from PORTC to PORTD. A switch is connected to External Interrupt pin to toggle LED 1. Timer0 and Timer 1 toggle LED 2 and LED3 after 10us and 20us respectively. The priority order is as follows

Timer1 >> External Interrupt >> Timer0

Marks: 4+

Question No. 5 (CLO No. 4)

(A) Construct a code in PIC Assembly Language to toggle PORTD 700 times.

7x100

(B) Create a traffic light system in which RED is on for 5m sec, Yellow is on for 2m sec and Green is on for 5m sec. There is a pedestrian button which when pressed will make the red light freeze for 10m sec and then the normal sequence continues for the traffic light.

RA1 → LED3.

RA0 → LED2.

RA3 → LED3.

20ms = (FFFF - 44XX + 1) 0.4μs 10MHz

10ms = (FFFF - 44XX + 1) 0.4μs

25 = (FFFF - 44XX + 1)

44XX = (65536 - 25)

65511 FF E7

0101	0101
0101	1010
0000	1010

20ms = (65536 - 44XX) 0.4μs

50 = 65486

FF CE